

SAI AKSHAY MENTA

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SUMMARY

AI/ML engineer with an MS in Artificial Intelligence from Northeastern, four peer-reviewed publications (ACL, IEEE, Springer), and applied experience at Genpact building Python-based data pipelines and retrieval systems for finance and operations workflows. Focus areas: Python, SQL, data pipelines, LLMs, and taking models from research into production on cloud (AWS/GCP).

EDUCATION

Northeastern University

Master of Science in Artificial Intelligence [GPA:3.8/4.0]

Sep 2024 - Apr 2026

Boston, MA

Coursework: Natural Language Processing, Large Language Models, Machine Learning, Advanced Perception

Amrita Vishwa Vidyapeetham University

Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) [GPA:3.7/4.0]

Oct 2020 - Jun 2024

Coimbatore, India

Coursework: Linear Algebra, Probability & Statistics, Advanced Reinforcement Learning, Big Data Analytics

SKILLS

Languages Python, SQL, JavaScript

Data Engineering ETL/Data Pipelines, Pandas, NumPy, PostgreSQL, ChromaDB, Power BI

ML/DL PyTorch, NLP, LLM Fine-tuning, Multimodal LLMs, Machine Learning, Computer Vision, MCP

Frameworks Hugging Face, LangChain, CrewAI, Scikit-learn, FastAPI, Flask, Streamlit, React, OpenCV

Cloud & Tools AWS, GCP, Docker, Kubernetes, Git, CI/CD, Ollama

EXPERIENCE

Research Assistant

Jan 2026 - Present

InsightX Lab

Boston, MA

- Built an end-to-end Python data pipeline that processes 2.1GB of Elliptic++ Bitcoin blockchain data into PyTorch Geometric graph objects, handling 1M+ nodes and 3.1M+ edges with temporal splits and class-weighted sampling to address 2% illicit class imbalance, completing in 37 seconds.
- Architected a multi-agent fraud detection system that combines GNN-based transaction and wallet scoring with rule-based compliance checks to produce auditable, human-readable compliance reports.

Data Scientist Intern

Feb 2024 - Jul 2024

Genpact

Hyderabad, India

- Built an embedding-based information retrieval system in Python to automate document search and response generation, improving response accuracy by 25% and cutting manual lookup time by 40%.
- Developed reliable Python data pipelines with validation, logging, and error handling across SQL data sources to support finance and operations workflows.
- Built interactive Power BI dashboards with SQL-driven data models for sales and pricing analysis, reducing reporting turnaround by 30% and enabling data-driven decisions across the team.

Research Assistant

Nov 2021 - Dec 2023

Amrita School of AI

Coimbatore, India

- Developed a multimodal reinforcement learning framework combining vision-text embeddings (CLIP) that outperformed PPO baselines; results accepted at ACL 2024 and featured in 4 peer-reviewed publications.
- Fine-tuned Whisper Large with LoRA (PEFT) to build a speech recognition pipeline, improving accuracy by 10% on dysarthric speech while reducing compute requirements.
- Built a medical diagnostic model achieving 84% accuracy, using diffusion models for synthetic data augmentation and Grad-CAM for interpretability.

PUBLICATIONS

- Improving Reinforcement Learning Agent Training using Text based Guidance, ACL Anthology [\[Link\]](#) Sep 2023
- A Few-Shot Approach to Dysarthric Speech Intelligibility Level Classification Using Transformers, 14th ICCNT [\[Link\]](#) Nov 2023
- Enhancing Knee Osteoarthritis Severity Level Classification Using Diffusion Augmented Images, Springer ICACECS [\[Link\]](#) Dec 2023
- Semi Supervised Flood Damage Detection Using Satellite Images, Springer Nature, ICCAIML [\[Link\]](#) Dec 2024

PROJECTS

Scaffold: AI-Powered Habit Coaching Platform

Jan 2026 - Apr 2026

Tech Stack: Next.js, React, TypeScript, Supabase, PostgreSQL, Groq LLM

[\[GitHub\]](#)

- Developed a full-stack AI coaching platform with a Supabase/PostgreSQL backend, designing SQL schemas and queries to personalize guidance to each user's personality and routine, integrated with Groq LLM inference across onboarding, planning, and feedback flows.
- In a 20-person comparative usability study, participants reported 19% higher satisfaction versus a generic baseline, validating the impact of personalized coaching on engagement.

InfoMeme: AI-Driven Meme Generation from News

Sep 2024 - Dec 2024

Tech Stack: Python, AWS (S3, EC2 GPU), OpenAI GPT-4 Vision, FAISS, Streamlit

[\[GitHub\]](#)

- Built an end-to-end multimodal data pipeline in Python combining LLM-based text generation with FAISS-powered semantic retrieval across 10,000+ news embeddings, cutting content matching latency by 35%.
- Automated the full workflow from news ingestion to meme output on AWS, using S3 for embedding and asset storage and EC2 GPU instances for inference, demonstrating hands-on experience with cloud-based agentic pipelines and scalable inference systems.

PromptPrint: Multi-Agent LLM Optimization System

MIT AI Hackathon – May 2025

Tech Stack: Python, Langchain, OpenRouter, LLM APIs, Energy Estimation APIs

[\[Demo\]](#)

- Built an agentic framework in Python that routes user queries across 15+ LLMs using reinforcement-based agent coordination, balancing latency, cost, and real-time carbon footprint scoring per request.
- Achieved 40% reduction in energy consumption versus a single-model baseline while maintaining response quality, demonstrating viable sustainability-aware LLM serving at the agentic layer.